Advanced Machine Learning, ETH Zurich

# Attention and transformers

*From one attention map to a trained transformer*

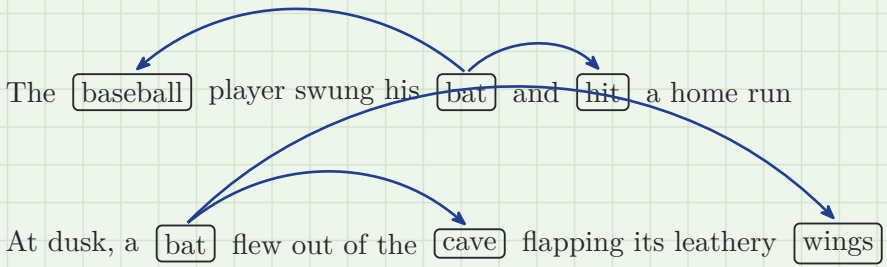
Carlos Cotrini

Autumn semester 2026

Entry 1

# Attention

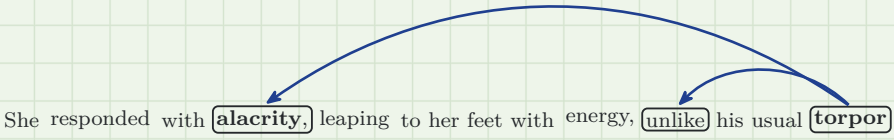
Words can find their meaning in a sentence by **attending to** other words!







- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?
- Alacrity: brisk and cheerful readiness.
- Torpor: a state of physical or mental inactivity; lethargy.


She responded with alacrity, leaping to her feet with energy, unlike his usual torpor in the mornings.

- We **define** a **value** representation for each word.
- We let other words **enrich** their representations by **attending** to other words.

- Wuggish, a toy language: every sentence describes at most three things.



*streept kat*



*bootet kat*

*Eight words*

| | |
|---|---|
| *kat* | cat |
| *snaak* | snake |
| *chaar* | chair |
| *streept* | striped |
| *bootet* | booted |
| *vennomus* | venomous |
| *wudden* | wooden |
| *brooken* | broken |



*vennomus snaak wudden chaar*



*streept snaak streept kat wudden chaar*

- Learn the eight words from pictures, then read bags of words in any order.

*wudden* *snaak* *vennomus* *chaar*



https://eth-cscs-hs26.github.io/CSCS-PAGE/viz/part1/wuggish/?dev=0

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?



*vennomus streept snaak wudden chaar*



*kat brooken chaar*

| | *kat* | | | *snaak* | | | *chaar* | | |
|---|---|---|---|---|---|---|---|---|---|
| | present? | *streept* | *bootet* | present? | *vennomus* | *streept* | present? | *wudden* | *brooken* |
| | 1, 0 | +1, −1 | +1, −1 | 1, 0 | +1, −1 | +1, −1 | 1, 0 | +1, −1 | +1, −1 |
| A | 0 | 0 | 0 | 1 | +1 | +1 | 1 | +1 | −1 |
| B | 1 | −1 | −1 | 0 | 0 | 0 | 1 | −1 | +1 |

## Query





## Keys

GreenChoice Supermarket



StoryStack Books



Threadline Clothing Co



Candy Cave



Spirit Cellar Liquors



## Values







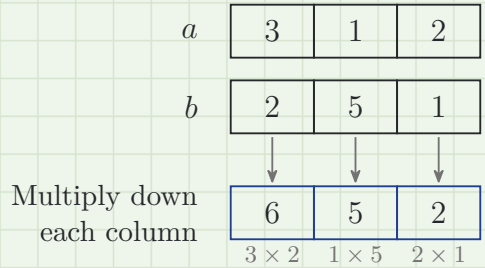






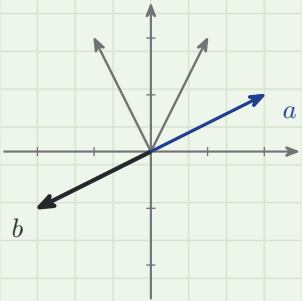

Entry 2

# The inner product




Then add $6 + 5 + 2 = \boxed{13}$ Inner product of $a$ and $b$

Inner product of $a$ and $b = a_1b_1 + a_2b_2 + a_3b_3$




$a = (2, 1)$

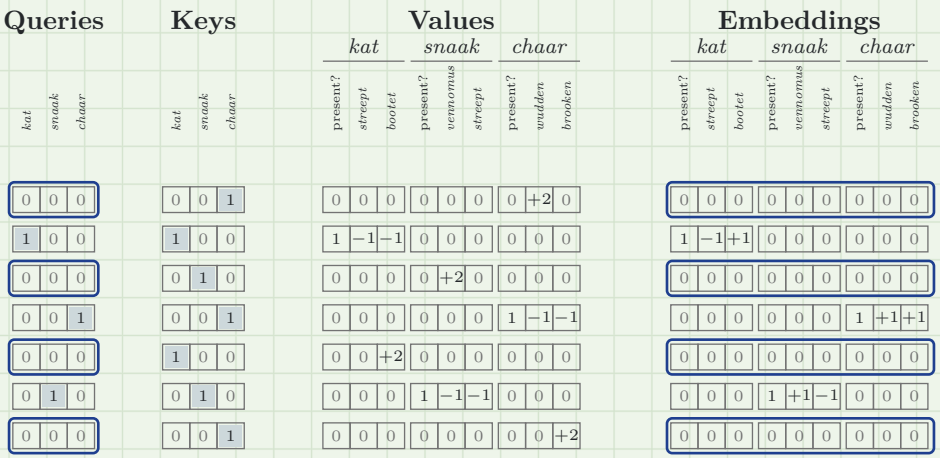
| | | |
|---|---|---|
| $b = (2, 1)$ | $2 \times 2 + 1 \times 1 = 5$ | Same direction |
| $b = (1, 2)$ | $2 \times 1 + 1 \times 2 = 4$ | Close directions |
| $b = (-1, 2)$ | $2 \times (-1) + 1 \times 2 = 0$ | At right angles |
| $b = (-2, -1)$ | $2 \times (-2) + 1 \times (-1) = -5$ | Opposite directions |

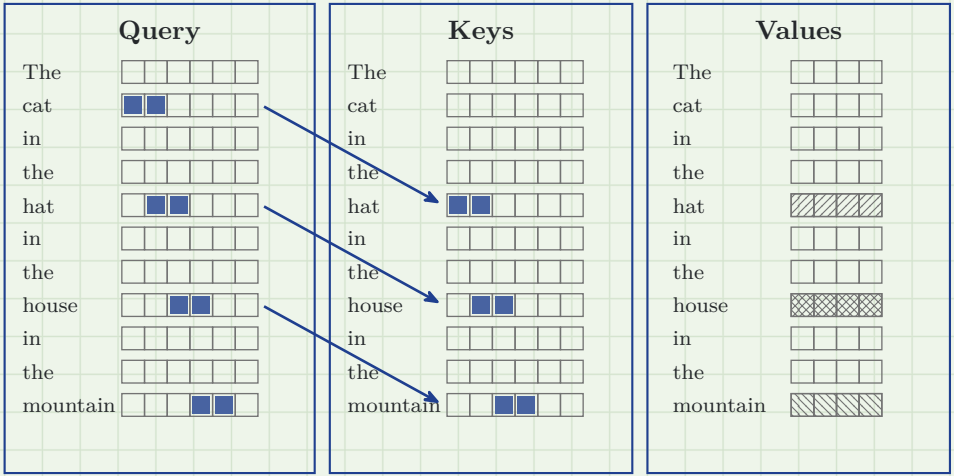
| | *kat* | *snaak* | *chaar* | Inner product |
|---|---|---|---|---|
| Query of *kat* | 1 | 0 | 0 | |
| Key of *kat* | 1 | 0 | 0 | **1** |
| Key of *bootet* | 1 | 0 | 0 | **1** |
| Key of *wudden* | 0 | 0 | 1 | **0** |
| Key of *streept* | 1 | 1 | 0 | **1** |

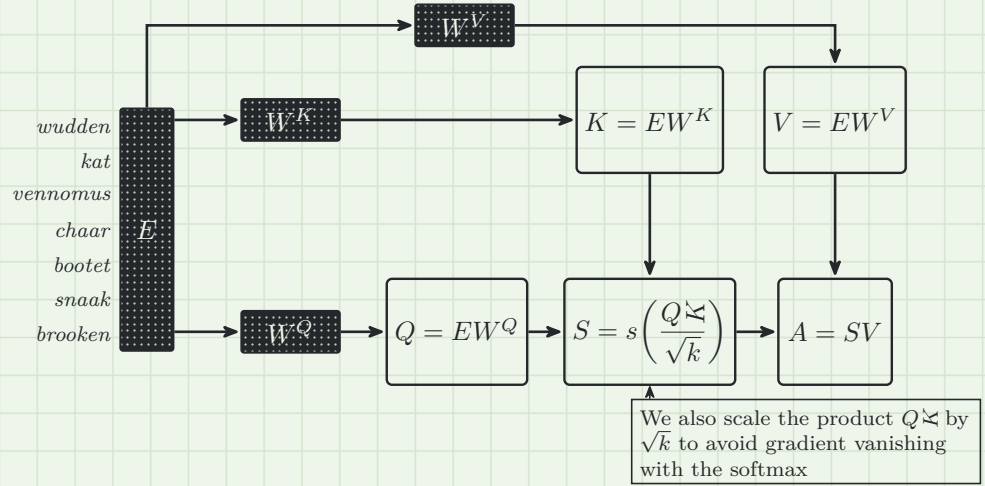
It counts the slots where both have a 1

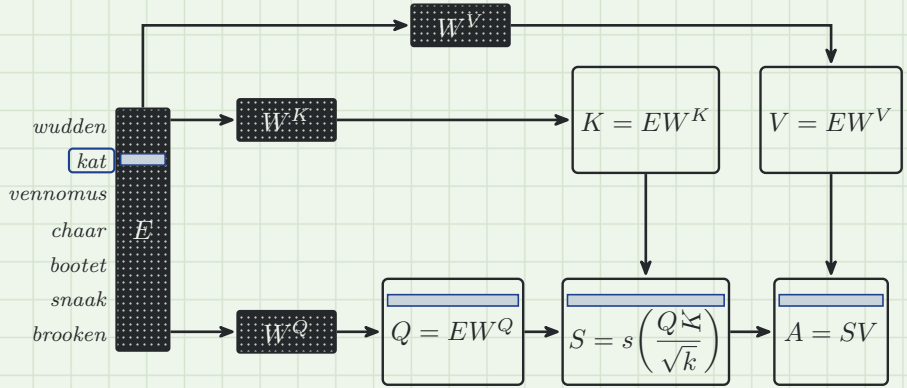
Entry 3

# The attention mechanism


Queries
Keys
Values
Embeddings
kat
snaak
chaar
present?
streept
bootet
vennomus
wudden
brooken
wudden
kat
vennomus
chaar
bootet
snaak
brooken


Query
Keys
Values
The
cat
in
the
hat
in
the
house
in
the
mountain


wudden
kat
vennomus
chaar
bootet
snaak
brooken
E
$W^V$
$W^K$
$W^Q$
$K = EW^K$
$V = EW^V$
$Q = EW^Q$
$S = s\left(\frac{Q\overleftarrow{K}}{\sqrt{k}}\right)$
$A = SV$
We also scale the product $Q\overleftarrow{K}$ by $\sqrt{k}$ to avoid gradient vanishing with the softmax


$W^V$
wudden
kat
vennomus
chaar
bootet
snaak
brooken
$E$
$W^K$
$K = EW^K$
$V = EW^V$
$W^Q$
$Q = EW^Q$
$S = s\left(\frac{QK}{\sqrt{k}}\right)$
$A = SV$

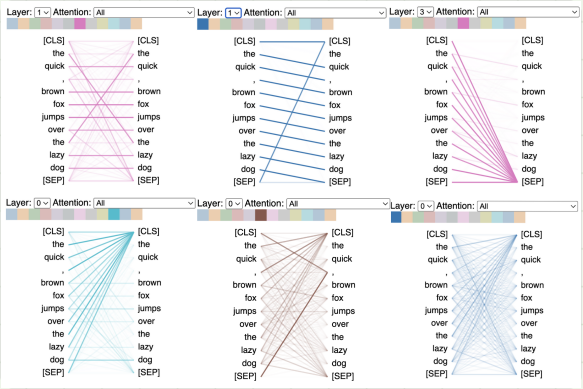
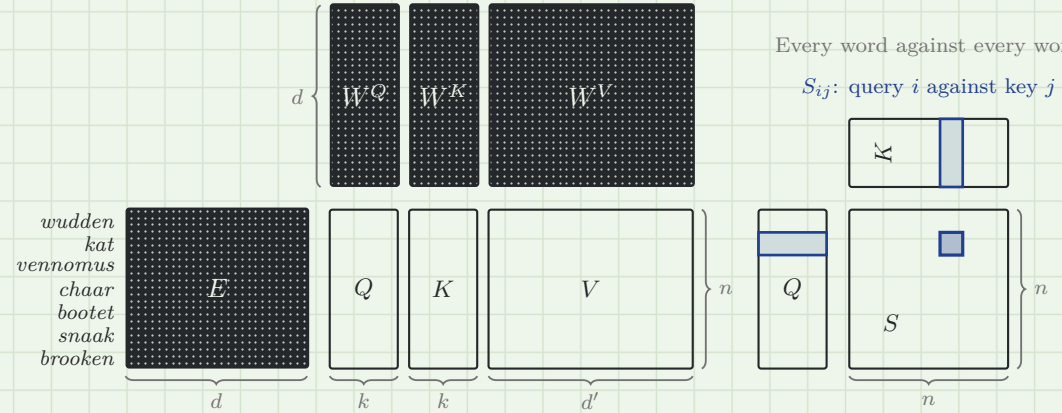


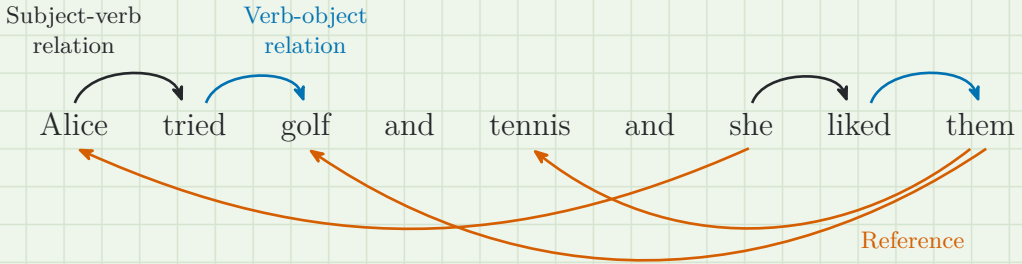

Image from: https://www.comet.com/site/blog/explainable-ai-for-transformers/

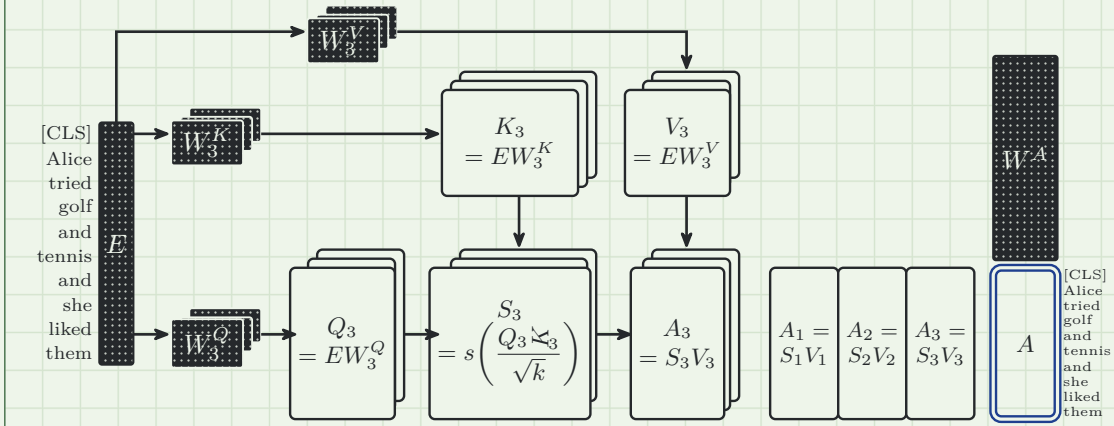

d
$W^Q$
$W^K$
$W^V$
Every word against every word
$S_{ij}$: query $i$ against key $j$
K
wudden
kat
vennomus
chaar
bootet
snaak
brooken
E
Q
K
V
n
Q
S
n
d
k
k
d′
n

Entry 4

# Multi-headed attention

Remember that we need multiple attention maps.

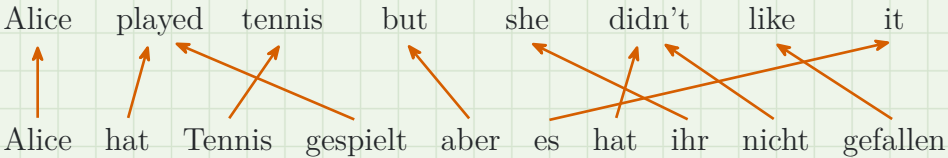


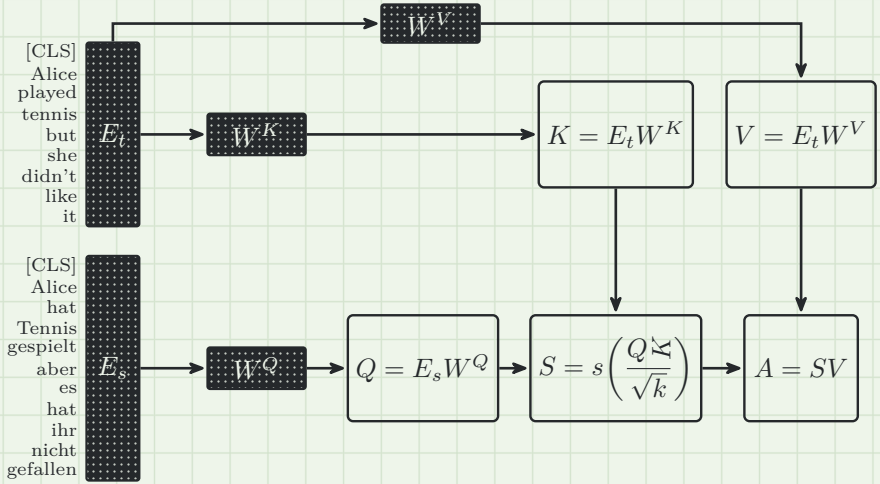

[CLS] Alice tried golf and tennis and she liked them
$E$
$W_3^V$
$W_3^K$
$W_3^Q$
$K_3 = EW_3^K$
$V_3 = EW_3^V$
$Q_3 = EW_3^Q$
$S_3 = s\left(\frac{Q_3 K_3}{\sqrt{k}}\right)$
$A_3 = S_3 V_3$
$A_1 = S_1 V_1$
$A_2 = S_2 V_2$
$A_3 = S_3 V_3$
$W^A$
$A$
[CLS] Alice tried golf and tennis and she liked them

Entry 5

# Cross-attention

- Consider the problem of translating from English to German.
- A different attention mechanism is needed.

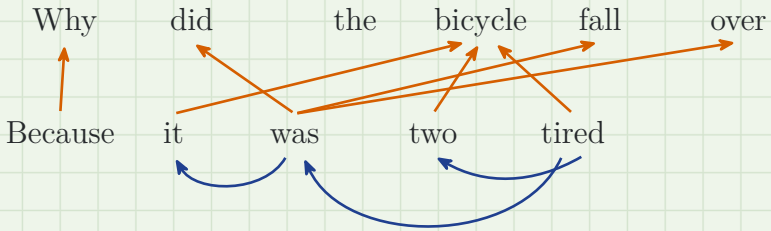



[CLS] Alice played tennis but she didn't like it
$E_t$
$W^V$
$W^K$
$K = E_t W^K$
$V = E_t W^V$
[CLS] Alice hat Tennis gespielt aber es hat ihr nicht gefallen
$E_s$
$W^Q$
$Q = E_s W^Q$
$S = s\left(\frac{Q\overline{K}}{\sqrt{k}}\right)$
$A = SV$

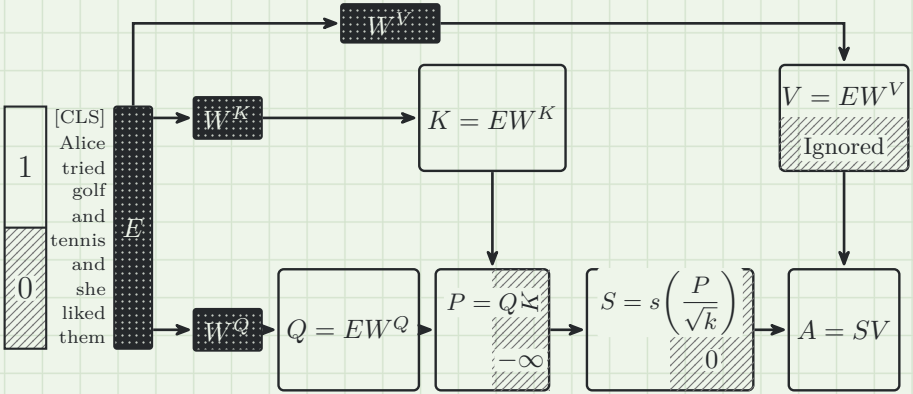
Entry 6

# Masked self-attention

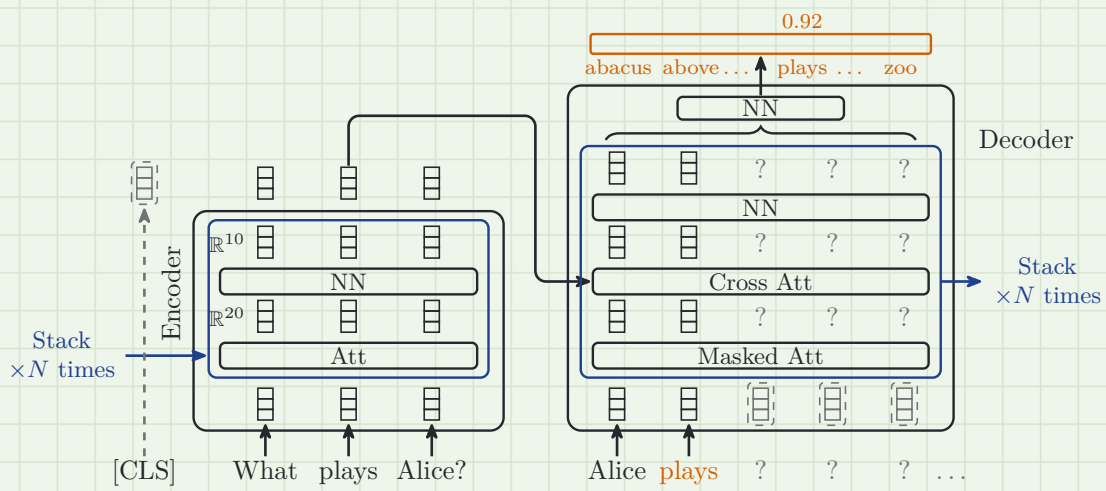
- Consider the task of generating text for answering questions.




For generating sentences, we need an attention mechanism that only looks at what has been produced so far.



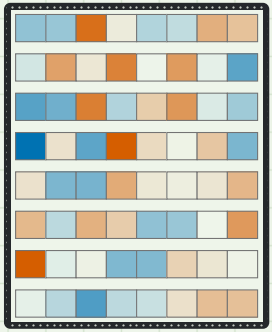

Note how the final embeddings depend only on the embeddings of the unmasked tokens!

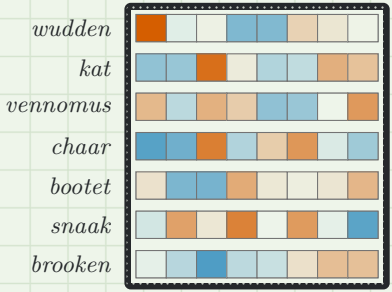

0.92
abacus above . . . plays . . . zoo
NN
Decoder
? ? ?
NN
? ? ?
Cross Att
? ? ?
Masked Att
Stack
×N times
Encoder
ℝ10
NN
ℝ20
Att
Stack
×N times
[CLS]
What plays Alice?
Alice plays ? ? ? . . .

Entry 7

# Training a transformer

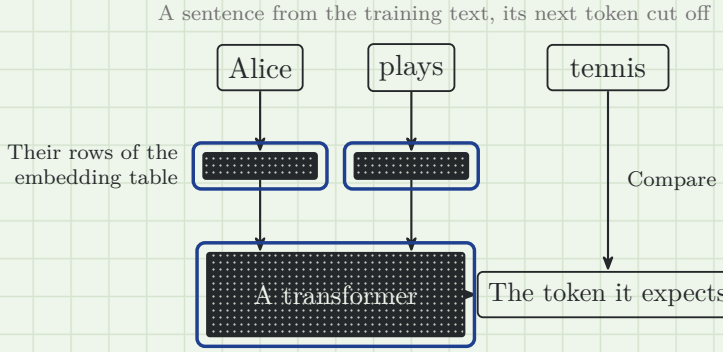
One row of numbers per word




$E$

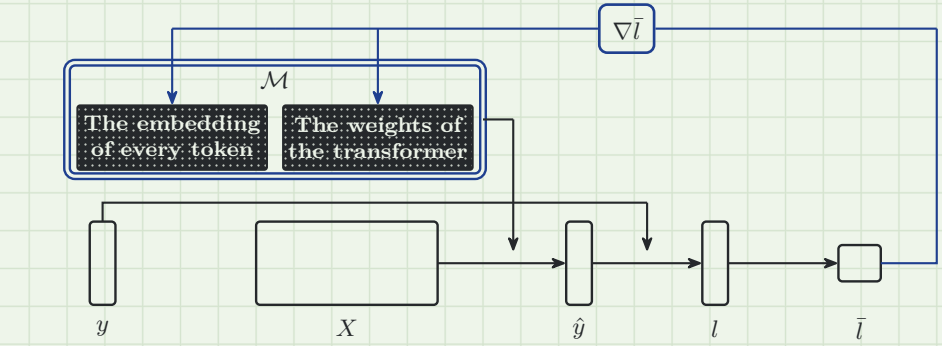
These numbers are weights too, like $W^Q$, $W^K$ and $W^V$. Training guesses them

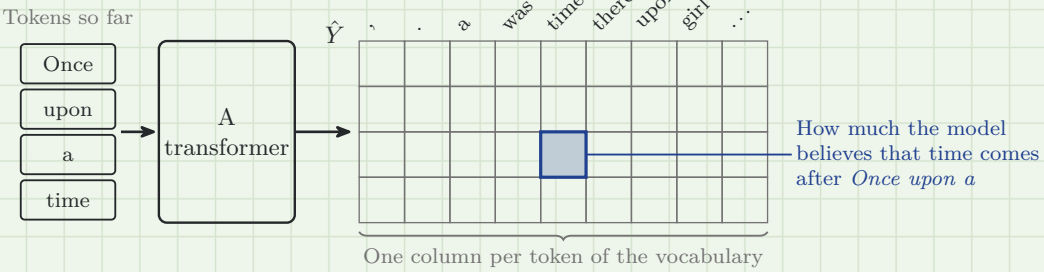


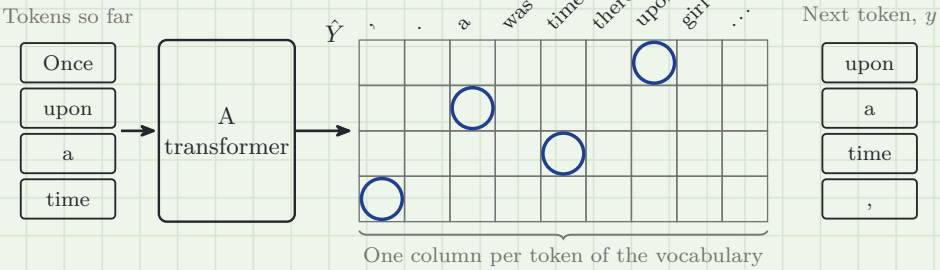

Not tennis? Every outlined weight moves a little toward it, both rows included

A row is text with its next token cut off. Every position of every document gives one.

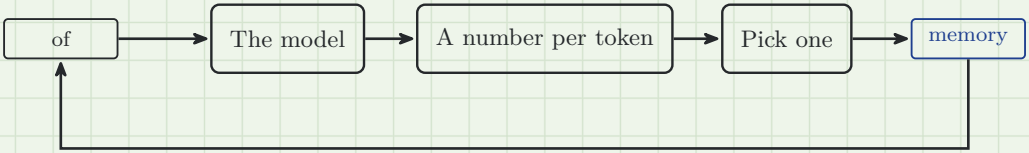
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.




Tokens so far
Once
upon
a
time
A transformer
$\hat{Y}$
,
.
a
was
time
there
upon
girl
...
How much the model believes that time comes after *Once upon a*
One column per token of the vocabulary




$$\bar{l} = -\tfrac{1}{n}\left(\log \hat{y}_{1,\text{upon}} + \log \hat{y}_{2,\text{a}} + \log \hat{y}_{3,\text{time}} + \cdots\right)$$


of
The model
A number per token
Pick one
memory
Appended to the prompt, and the whole thing runs again

Entry 8

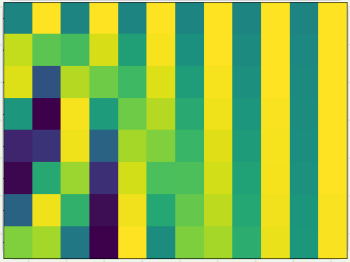
# Positional encodings

- Consider the sentences *the dog chased the cat* and *the cat chased the dog*.
- They are permutations of each other, so the attention maps will also just be permutations of each other.
- Note that attention mechanisms do not pay attention to the position of a token in the sentence.

| | | | | | | |
|---|---|---|---|---|---|---|
| [CLS] | 0 | 0 | 0 | 0 | 0 | 0 |
| because | 1 | 0 | 0 | 0 | 0 | 0 |
| it | 0 | 1 | 0 | 0 | 0 | 0 |
| was | 1 | 1 | 0 | 0 | 0 | 0 |
| two | 0 | 0 | 1 | 0 | 0 | 0 |
| tired | 1 | 0 | 1 | 0 | 0 | 0 |
| [EOS] | 0 | 1 | 1 | 0 | 0 | 0 |

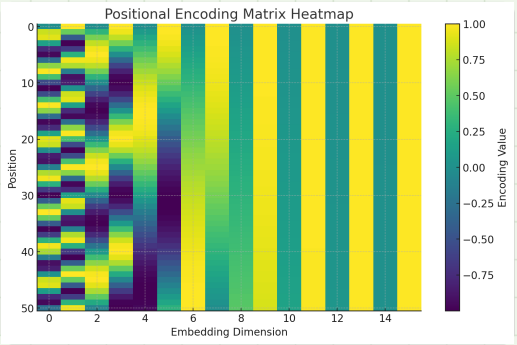
$BP$

$$BP_{ij} = \left(i \div 2^j\right) \bmod 2$$
$$= \lfloor 2^{-j} i \rfloor \bmod 2$$



- We actually use both sin and cos and then interleave them.

- The positional encoding matrix $p$ has $T$ rows and $d$ columns.
- The value $T$ is the maximum number of tokens in a sentence.
- The value $d$ is the dimensionality of the positional embedding.
- The encoding of token $i$ is defined through the formulas
  $p_{i,2j} = \sin\left(\alpha^{-2j} i\right)$ and
  $p_{i,2j+1} = \cos\left(\alpha^{-2j} i\right)$,
  for $j \leq d/2$ and $\alpha = 10^{4/d}$.



- Concatenation substantially increases the dimensionality of the embedding, increasing the number of parameters to be trained.
- With sufficient dimensionality, since the positional embeddings are somewhat statistically independent from the token embeddings, both embeddings are somewhat orthogonal. Hence, the addition still preserves information about the two vectors.
- Since most of the operations done in an attention mechanism are linear, the final embedding is somewhat the addition of the result of the operations applied to both the positional and the token embeddings.

Entry 9

# Residual blocks

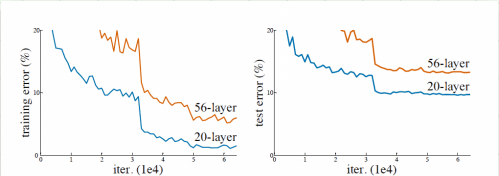



Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Figure 1 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.
`https://arxiv.org/abs/1512.03385`

- When deeper networks are able to start converging, a *degradation problem* appeared. Training performance degrades rapidly. This is not because of overfitting.
- The problem is mostly due to difficulties of information propagating, both forward and backward, through the layers in the network.

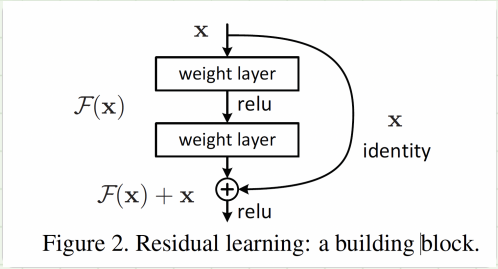



Figure 2. Residual learning: a building block.

Figure 2 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.
https://arxiv.org/abs/1512.03385

- To address this problem, residual blocks were introduced. They work by adding the output of a layer to the output of a few layers ahead.

Entry 10

# Transformer architecture

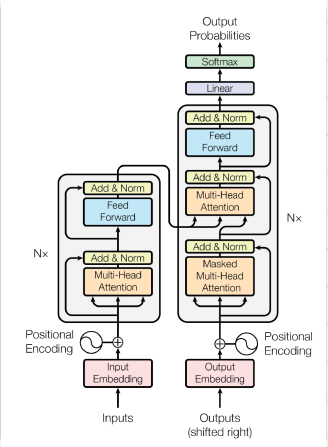



Figure 1 of

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser and Illia Polosukhin.

*Attention Is All You Need.*

Advances in Neural Information Processing Systems 30 (NeurIPS 2017).

`https://arxiv.org/abs/1706.03762`